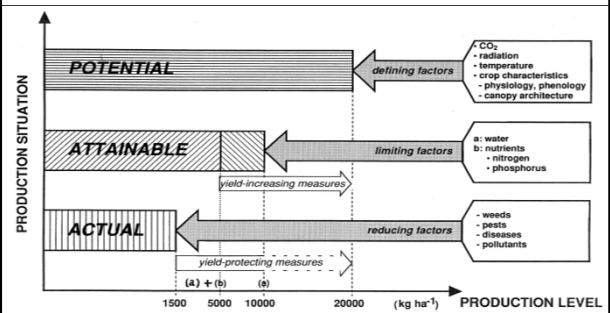


Lintul1: Potential crop growth

Dr. ir. Peter Leffelaar - Wageningen University



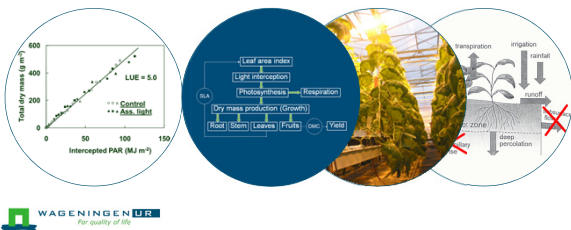
Potential, attainable and actual production levels



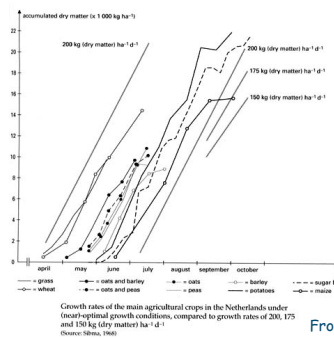
From: Lovenstein, Lantinga, Rabbinge and van Keulen, 1995

Lintul1: Potential crop growth

Introduction

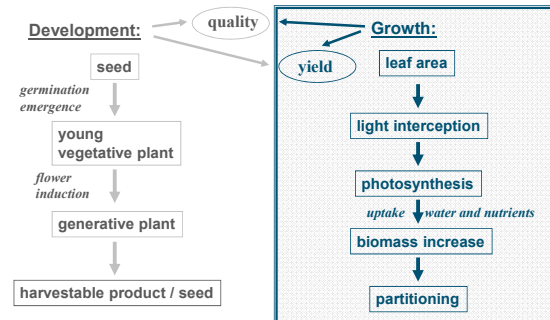


Optimal growth rates - rule of thumb



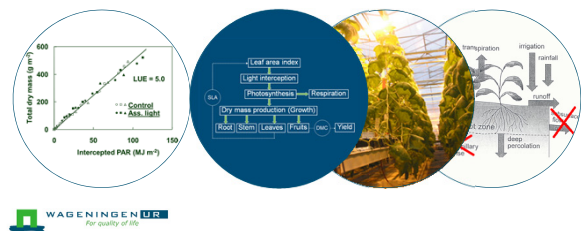
From: Sibma, 1968

Lintul1: Potential crop growth - scope of the course -

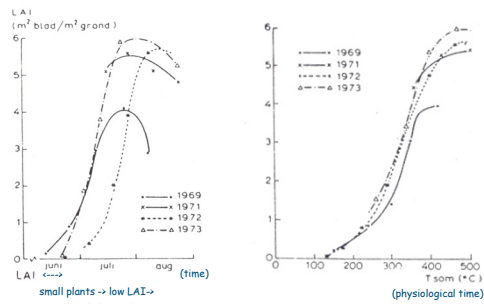


Lintul1: Potential crop growth

Light and light interception



Leaf area of maize as a function of (physiological) time



From: Sibma, 1987

Spectral composition of global radiation

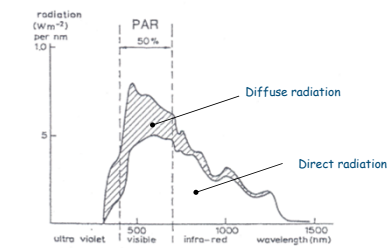
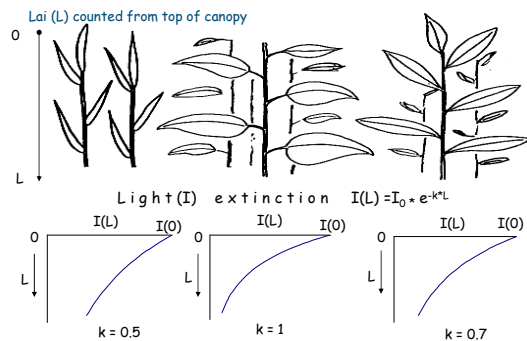


Figure 3.2 Spectral composition of global radiation, discriminating between direct radiation (clear area) and diffuse (i.e. indirect) radiation scattered by clouds (hatched area). PAR = photosynthetically active radiation. (Source: Monteith and Unsworth, 1990, Principles of environmental physics, Edward Arnold, London, 291 p.)

From: Monteith & Unsworth, 1990

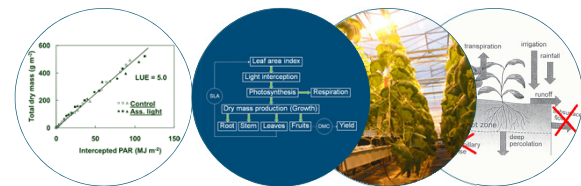
Light extinction is a function of LAI



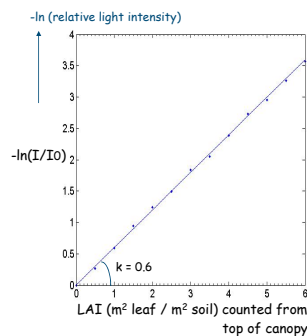
From: Coombs et al., 1985

Lintul1: Potential crop growth

Light (or Radiation) use efficiency



Light extinction coefficient, k, follows from slope



$$I(L) = I_0 \cdot e^{-k \cdot L}$$

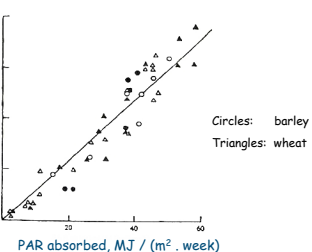
$$\Rightarrow -\ln\left(\frac{I}{I_0}\right) = k \cdot L$$

maize

From: Sibma, 1987

Light Use Efficiency, LUE, follows from slope

Crop growth rate
 $g / (m^2 \cdot week)$

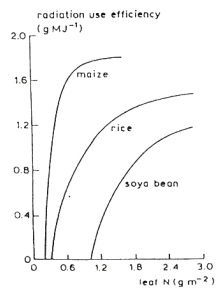


PARAM LUE = 3.0 (g Dry Matter per MJ PAR)
LUE Pinus = 1.9
LUE Quercus = 2.1

From: Gallagher & Biscoe, 1978

See for an excellent discussion about LUE:
Sinclair and Muchow, 1999.

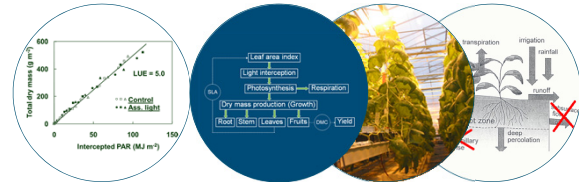
Radiation use efficiency as a function of leaf nitrogen content per unit area



From: Sinclair & Horie, 1989

Lintul1: Potential crop growth

Crop development

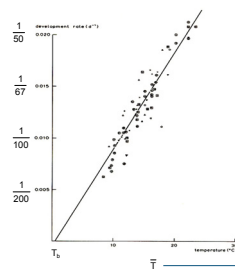


Lintul1: Potential crop growth

Light interception, utilization and growth



Development rate of spring wheat as a function of temperature from emergence till flowering . . .



$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.0009 \Rightarrow$$

$$\text{"temperature sum"} = \bar{T} \cdot \# \text{ days} \Rightarrow$$

$$\frac{1}{0.0009} = 1110 \text{ }^{\circ}\text{Cd}$$

Data from over 20 papers assembled by: Van Keulen & Seligman, 1987

Light interception, utilization and growth in LINTUL

TITLE LINTUL

INITIAL

INCON WTOTI = 5.

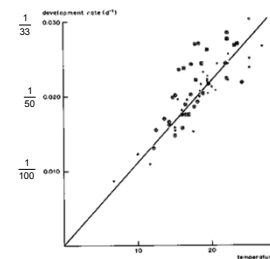
TIMER STTIME=1; FINTIM=100.; DELT=1.; PRDEL = 5.
TRANSLATION_GENERAL DRIVER = 'EUDRIV'
PRINT LAI, WTHA, PARINT

DYNAMIC
LAI = WTOTAL * LWR * SLA
PARINT = (0.5 * DTR) * (1 - EXP(-K * LAI))
L = W / F₀ S₀ compare Eq.(13)
I_{int} = 0.5 DTR (1 - e^{-K}) Eq.(2)
GTOTAL = LUE * PARINT
WTOTAL = INTGRL(WTOTI, GTOTAL)
WTHA = WTOTAL / 100.
dW/dt = RUE I_{int} Eq.(3)
W = ∫ dW/dt dt Eq.(4)

PARAM DTR = 20.; LWR = 0.5; SLA = 0.022; LUE = 3.0; K = 0.6

END
STOP
ENDJOB

. . . and from flowering to ripeness



$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.00103 \Rightarrow$$

$$\bar{T} \cdot \# \text{ days} = \frac{1}{0.00103} = 970 \text{ }^{\circ}\text{Cd}$$

Vegetative Phase: 1110 °Cd
Generative Phase: 970 °Cd
Whole growing season: 2080 °Cd

finish condition Lintul

Data from over 20 papers assembled by: Van Keulen & Seligman, 1987

Use of weather data in LINTUL

TITLE LINTUL

* 1. Initial conditions and run control
FINISH TSUM > 2080.

* 2. Environmental data and temperature sum
WEATHER WTRDIR=C:\SYS\WEATHER\; CNTR=IT; ISTN=3; ...
IYEAR=1985

DTR = RDD / 1.E+6

DAVTMP = 0.5 * (TMMN + TMMX)

DTEFF = MAX(0., DAVTMP - TBASE)

EMERG = INSW(TIME-DOYEM, 0., 1.)

TSUM = INTGRL(ZERO, RTSUM)

RTSUM = DTEFF * EMERG

$$T_{sum} = \int (T_a - T_b) dt$$

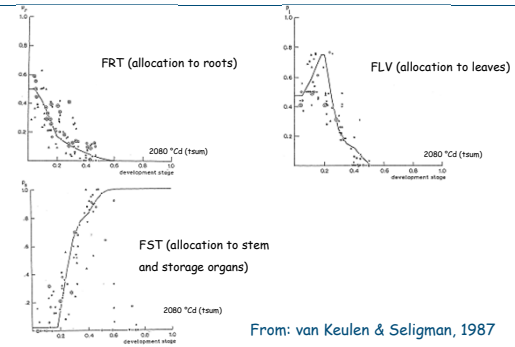
* 6. Functions and parameters for spring wheat

PARAM TBASE = 0.

PARAM DOYEM = 32.



Substrate allocation patterns in Spring Wheat: Measured

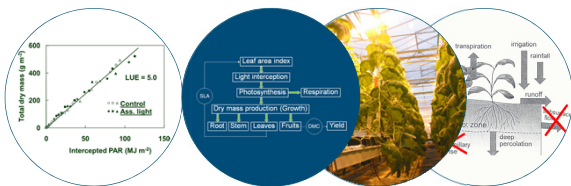


From: van Keulen & Seligman, 1987

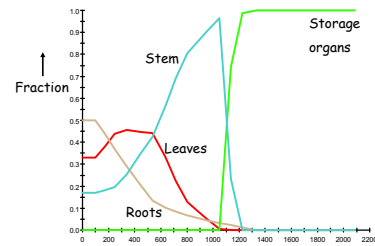


Lintul1: Potential crop growth

Development stages and allocation of dry matter



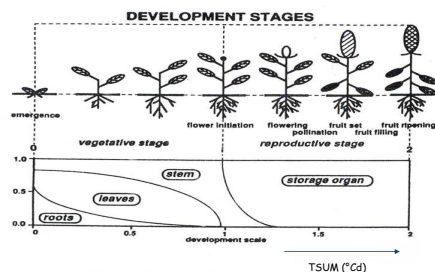
Substrate allocation patterns in Spring Wheat: In model



Allocation pattern of assimilates for spring wheat
as used in Lintul (Light interception and utilization)



Crop development stages



From: L6venstein, Lantinga, Rabbinge and van Keulen, 1995



Growth rates and dry matter production of plant organs

TITLE LINTUL

FRT = AFGEN(FRTTB, TSUM)

FLV = AFGEN(FLVTB, TSUM)

FST = AFGEN(FSTTB, TSUM)

FSO = AFGEN(FSOTB, TSUM)

WLVG = INTGRL(WLVI, RWLVG)

WLVD = INTGRL(ZERO, DLV)

WST = INTGRL(ZERO, RWST)

WSO = INTGRL(ZERO, RWST)

WSOHA = WSO / 100.

WRT = INTGRL(ZERO, RWRT)

WLW = WLVG + WLVD

RWLVG = GTOTAL * FLV - DLV

RWST = GTOTAL * FST

RWSO = GTOTAL * FSO

RWRT = GTOTAL * FRT

$$dW_{lv} / dt = RUE I_{int} F_{lv} - r_d W_{lv} \quad \text{Eq.(8)}$$

$$dW_{st} / dt = RUE I_{int} F_{st} \quad \text{Eq.(7)}$$

$$dW_{so} / dt = RUE I_{int} F_{so} \quad \text{Eq.(7)}$$

$$dW_{rt} / dt = RUE I_{int} F_{rt} \quad \text{Eq.(7)}$$



Functions for spring wheat

```

TITLE LINTUL

FUNCTION FRITB = 0,0.50 , 110., 0.50 , ...
275.,0.34 , 555.,0.12 , 780.,0.07 , ...
1055.,0.03 , 1160.,0.02 , 1305.,0. , 2500.,0.

FUNCTION FLVTB = 0,0.33 , 110.,0.33 , ...
275.,0.46 , 555.,0.44 , 780.,0.14 , ...
1055.,0. , 2500.,0.

FUNCTION FSTTB = 0,0.17 , 110.,0.17 , ...
275.,0.20 , 555.,0.44 , 780.,0.79 , ...
1055.,0.97 , 1160.,0. , 2500.,0.

FUNCTION FSOTB = 0,0. , 110.,0. , ...
1055.,0. , 1160.,0.98 , 1305.,1. , 2500.,1.
    
```

Parameters for spring wheat & Growth of Leaf Area (GLA)

```

PARAM LAII = 0.012 ; SLA = 0.022 ; RGRL = 0.009 ; ...
TSUMAN = 1110 ; LAICR = 4 ; RDRSHM = 0.03
FUNCTION RDRT = -10,0.03 , 10,0.03 , 15,0.04 , 30,0.09 , 50,0.09
*-----
SUBROUTINE GLA(TIME,DOYEM,DTEFF,TSUM,LAII,RGRL,DELT,
$ SLA ,LAI,GLV,
IMPLICIT REAL (A-Z)
GLAI)

*---- Growth during maturation stage:
GLAI = SLA * GLV dL/dt = SL Fh RUE Iint Eq(7 & 13)

*---- Growth during juvenile stage:
IF ((TSUM.LT.330.) .AND. (LAI.LT.0.75))
$ GLAI = LAI * (EXP(RGRL * DTEFF * DELT) - 1.) / DELT Eq(12)

*---- Growth at day of seedling emergence:
IF ((TIME.GE.DOYEM) .AND. (LAI.EQ.0.))
$ GLAI = LAII / DELT

*---- Growth before seedling emergence:
IF (TIME.LT.DOYEM) GLAI = 0.

RETURN
END
    
```

Lintul1: Potential crop growth

Early leaf growth

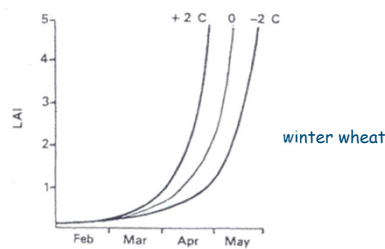


Lintul1: Potential crop growth

Summarizing relational diagram and meaning symbols

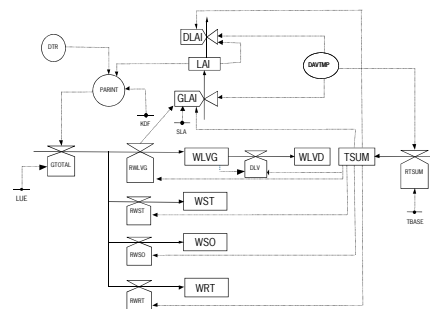


Early leaf growth: temperature dependent



From: Monteith, 1966

Relational diagram of Lintul1



Symbols are explained in next slides

Forrester notation . . .

Symbol		Dimension
	Integral	Amount
	Differential equation (decision about rate)	Amount/time
	Flow of substance	Amount/time
	Discontinuous flow of substance	Amount/time
	Flow of information	No unit
	Input variable	Not known beforehand
	Auxiliary variable	Not known beforehand

Practical exercises Lintul1

Exercises on the following subjects:

- Consequence of weather and different emergence dates on crop growth
- The sensitivity of yield for some important crop parameters or, differently stated, the consequence of uncertainties in parameter values
- Effects of parameter changes in light extinction (light capture), and light use efficiency on growth
- Potential yield as a function of emergence dates

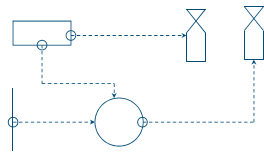
There are data provided for a dry, an average and a wet year

Choose between Brazil and the Netherlands

Prepare a report for yourself on the PC to prepare for the exam

. . . Forrester notation (continued)

Information flows can only be used:
Information flows should point towards a place of decision, never to a state:



To be able to do the exercises, study:

- Theory: "Lintul-1: potential crop growth"
- This power point presentation

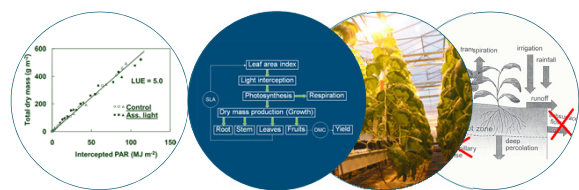
Lintul1: Potential crop growth

Exercises



Lintul1: Potential crop growth

References



References used in the slides

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